Water Security Plan Report

Date : 29-09-2023 Time : 5:10 PM User Name : patotlani@gmail.com

STATE: GUJARAT_24 » DISTRICT: BANASKANTHA_441 » BLOCK: DANTIWADA_3735 » GRAM PANCHAYAT: AKOLI_155943 » YEAR: 2023-24 »

Gram Panchayat Information

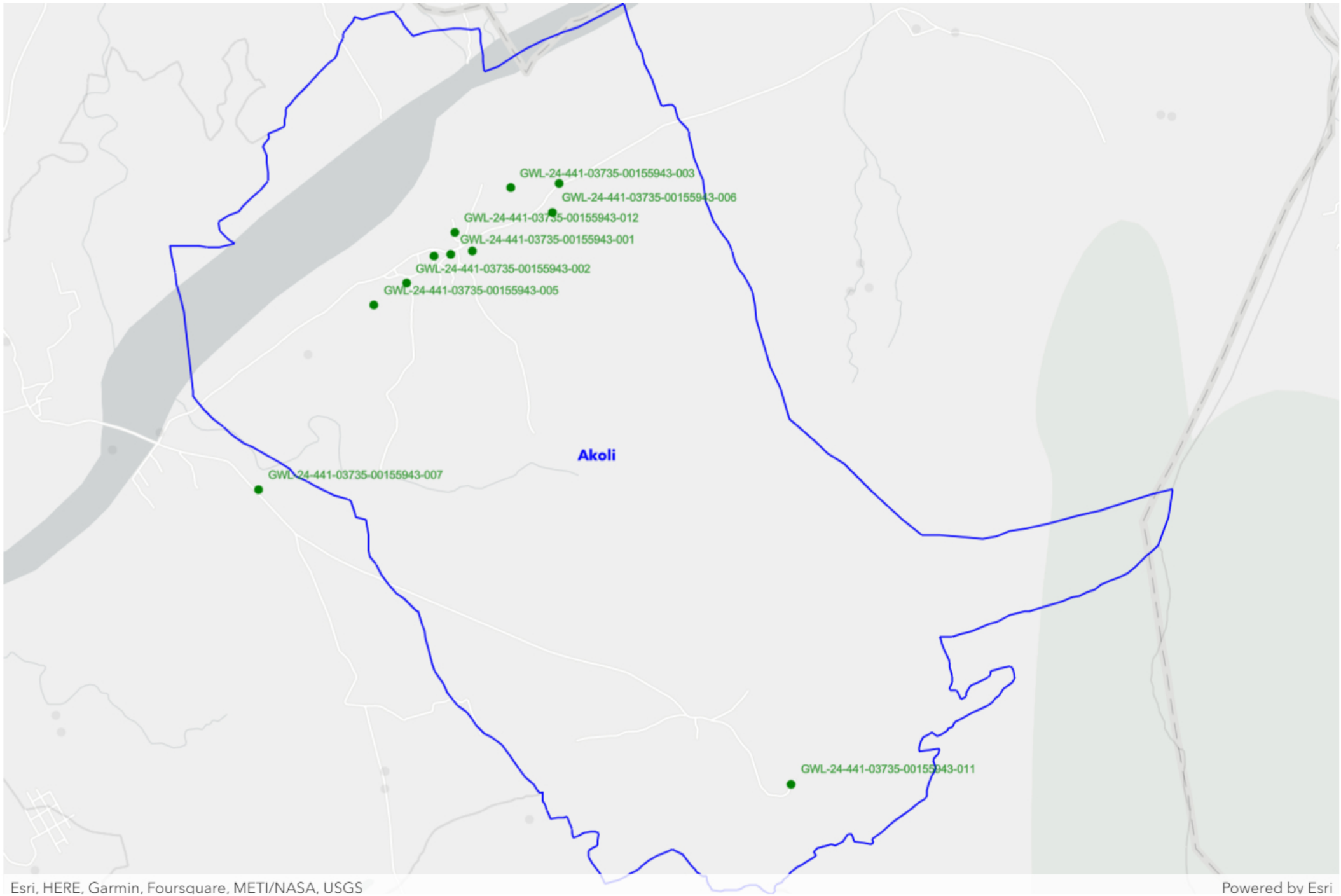
State:	Gujarat_24	District:	Banaskantha_441
Block / Taluk :	Dantiwada_3735	GP Name :	Akoli_155943
GP ID :	155943	Block Area (Ha) :	45250.16
GP area (Ha) :	1891.76	Watershed Code :	SRWT003
Watershed name :	SARASWATI	Watershed area in Block / Taluk (Ha) :	131645
Sub-basin Code :	5B	Sub basin name :	Saraswati
Basin Code :	20	Basin Name :	Kutch

Location Map

Watershed Map

Aquifer Map

Ground Water Level

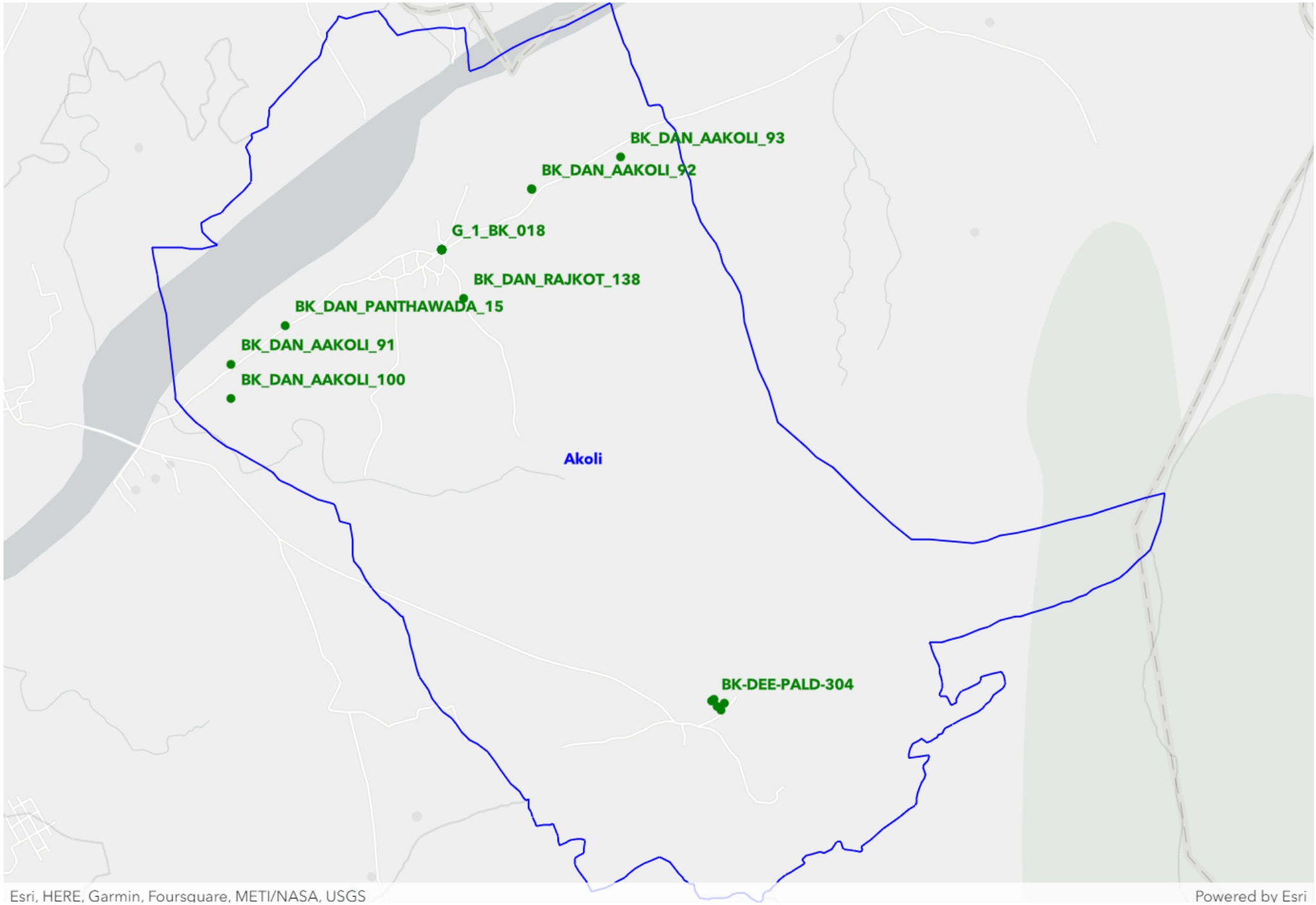


S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
1	Akoli	Jetabhai lakhabhai rabari	24.49656	72.38348	07-06-23 1	155943-001	Tube Well	null	Private	null	0.2	230	1	0	07-06-23
	Akoli	Jetabhai lakhabhai rabari	24.49656	72.38348	07-06-23 1	155943-001	Tube Well	null	Private	null	0.2	230	1	0	07-06-23
2	Akoli	Godabhai bhagabhai rabari	24.4948	72.3805	07-06-23 1	155943-002	Tube Well	null	Private	null	0.2	260	1	90	07-06-23
	Akoli	Godabhai bhagabhai rabari	24.4948	72.3805	07-06-23 1	155943-002	Tube Well	null	Private	null	0.2	260	1	0	07-06-23
	Akoli	Godabhai bhagabhai rabari	24.4948	72.3805	07-06-23 1	155943-002	Tube Well	null	Private	null	0.2	260	1	89.75	06-09-23
3	Akoli	Amraji nethiji rabari	24.50066	72.38753	07-06-23 1	155943-003	Tube Well	null	Private	null	0.2	200.25	1	97.18	06-09-23
	Akoli	Amraji nethiji rabari	24.50066	72.38753	07-06-23 1	155943-003	Tube Well	null	Private	null	0.2	200.25	1	98	07-06-23
	Akoli	Amraji nethiji rabari	24.50066	72.38753	07-06-23 1	155943-003	Tube Well	null	Private	null	0.2	200.25	1	0	07-06-23
4	Akoli	Khegarji nethiji rabari	24.49645	72.38235	07-06-23 1	155943-004	Tube Well	null	Private	null	0.2	260.6	1	95	07-06-23
	Akoli	Khegarji nethiji rabari	24.49645	72.38235	07-06-23 1	155943-004	Tube Well	null	Private	null	0.2	260.6	1	0	07-06-23
	Akoli	Khegarji nethiji rabari	24.49645	72.38235	07-06-23 1	155943-004	Tube Well	null	Private	null	0.2	260.6	1	92.85	06-09-23

S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
5	Akoli	Patel nanjibhai motibhai	24.49346	72.37828	07-06-23 1	155943-005	Tube Well	null	Private	null	0.2	140.8	0.3	40.5	07-06-23
	Akoli	Patel nanjibhai motibhai	24.49346	72.37828	07-06-23 1	155943-005	Tube Well	null	Private	null	0.2	140.8	0.3	0	07-06-23
	Akoli	Patel nanjibhai motibhai	24.49346	72.37828	07-06-23 1	155943-005	Tube Well	null	Private	null	0.2	140.8	0.3	38.65	06-09-23
6	Akoli	Girishbhai nanjibhai patel	24.49913	72.39034	07-06-23 1	155943-006	Tube Well	null	Private	null	0.2	160.9	0.1	86.42	06-09-23
	Akoli	Girishbhai nanjibhai patel	24.49913	72.39034	07-06-23 1	155943-006	Tube Well	null	Private	null	0.2	160.9	0.1	91.78	07-06-23
	Akoli	Girishbhai nanjibhai patel	24.49913	72.39034	07-06-23 1	155943-006	Tube Well	null	Private	null	0.2	160.9	0.1	91.78	07-06-23
7	Akoli	Kalabhai motibhai patel	24.48212	72.37051	07-06-23 1	155943-007	Tube Well	null	Private	null	0.2	190.2	1	58.25	06-09-23
	Akoli	Kalabhai motibhai patel	24.48212	72.37051	07-06-23 1	155943-007	Tube Well	null	Private	null	0.2	190.2	1	60.3	07-06-23
8	Akoli	Nareshbhai kalabhai patel	24.50091	72.39079	07-06-23 1	155943-008	Tube Well	null	Private	null	0.2	190.2	0.2	40.8	07-06-23
	Akoli	Nareshbhai kalabhai patel	24.50091	72.39079	07-06-23 1	155943-008	Tube Well	null	Private	null	0.2	190.2	0.2	39.41	06-09-23
9	Akoli	Fulshi karanshi thakor	24.49677	72.38494	07-06-23 1	155943-009	Tube Well	null	Private	null	0.2	220.4	1	33.49	06-09-23

S.No.	Site Name	Location Details	Latitude	Longitude	Date	Well ID	Type of Well	Other Well Type	Ownership Type	Government Source	Diameter of well(m)	Depth of Well(mbmp)	Height of Measuring Point (mbgl)	Water Level (mbgl)	Water Level Date
	Akoli	Fulshi karanshi thakor	24.49677	72.38494	07-06-23 1	155943-009	Tube Well	null	Private	null	0.2	220.4	1	35.49	07-06-23
10	Akoli	Maharshi bhamarshi thakor	24.54017	72.35497	07-06-23 1	155943-010	Tube Well	null	Private	null	0.2	170.45	1	29.4	07-06-23
	Akoli	Maharshi bhamarshi thakor	24.54017	72.35497	07-06-23 1	155943-010	Tube Well	null	Private	null	0.2	170.45	1	26.85	06-09-23
11	Akoli	Ganguvada mahadev mandir baju ma	24.464035	72.40645	02:36.2	155943-011	Piezometer	null	Government	SGWB	0.162	60	1	22.72	13-09-23
12	Akoli	Sankarbhagvan na mandir ma	24.497908	72.38376	57:29.9	155943-012	Piezometer	null	Government	SGWB	0.162	90	1	21.78	15-09-23

Ground Water Quality

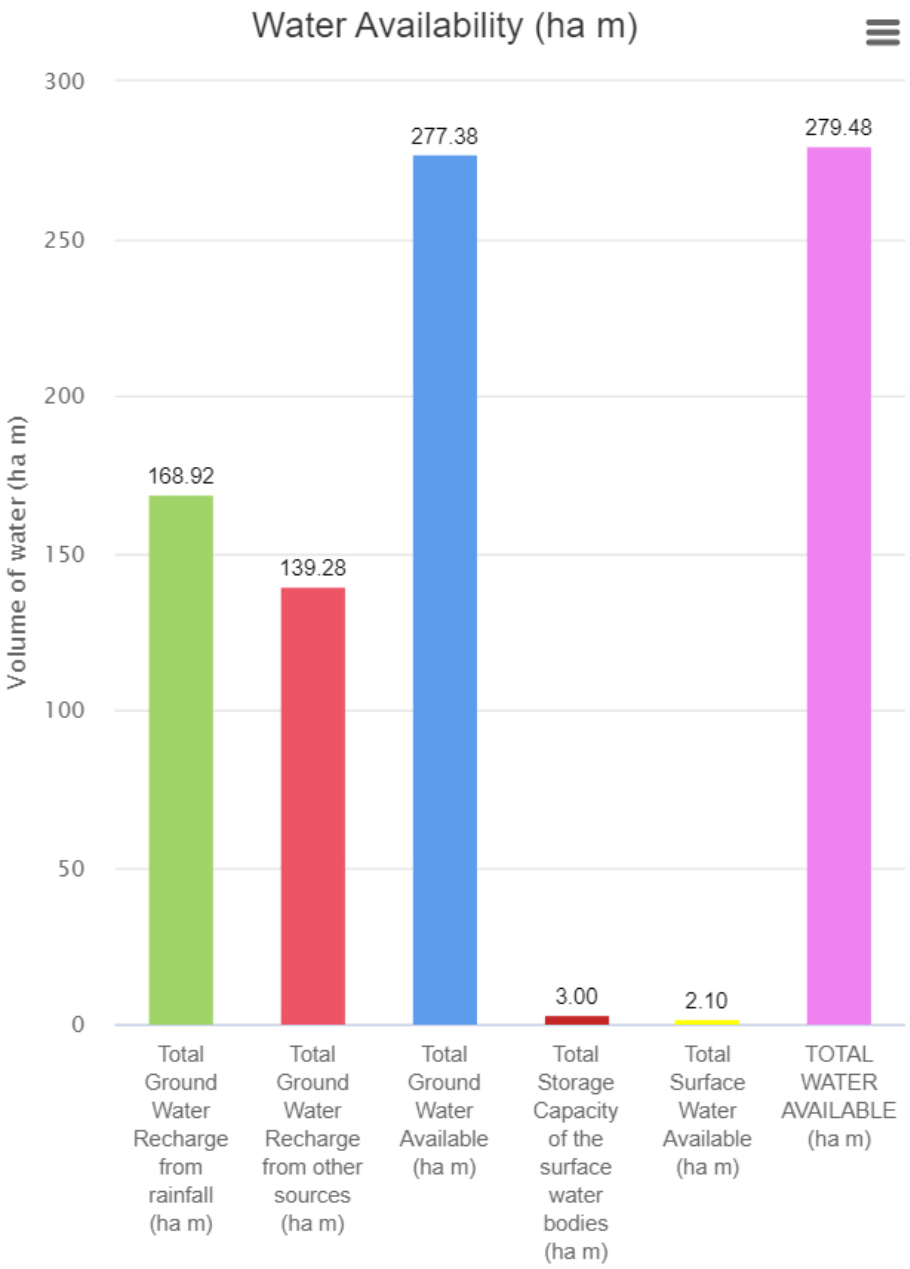


S.No.	Village Name	Location	Latitude	Longitude	Year	Well Type	Source	Well ID
1	PANSVAL	Devda Hadmatsingh Jethusingh	24.487897	72.369799	2022	Tube Well	ATAL JAL	BK_DAN_AAKOLI_100
2	AAKOLI	Mali Khemaji Bhikhaji	24.489966	72.369799	2022	Tube Well	ATAL JAL	BK_DAN_AAKOLI_91
3	AAKOLI	mali Sonaji Bhikhaji	24.500638	72.389881	2022	Tube Well	ATAL JAL	BK_DAN_AAKOLI_92
4	AAKOLI	mali Dahyabhai Bhutaji	24.502571	72.395841	2022	Tube Well	ATAL JAL	BK_DAN_AAKOLI_93
5	AAKOLI	mali Menaji Monji	24.5006	72.3899	2022	Tube Well	ATAL JAL	BK_DAN_AAKOLI_94
6	PANTHAWADA	Chowdhury Virabhai Premabhai	24.492322	72.373425	2022	Tube Well	ATAL JAL	BK_DAN_PANTHAWADA_15
7	RAJKOT	Thakor Virchandji Chamanbhai	24.493984	72.385338	2022	Tube Well	ATAL JAL	BK_DAN_RAJKOT_138
8	PALDI	PARBATBHAI AMBABHAI CHUHAN	24.469524	72.401922	2022	Tube Well	ATAL JAL	BK-DEE-PALD-304
9	PALDI	REMESHBHAI MALABHAI CHUHAN	24.469623	72.402069	2022	Tube Well	ATAL JAL	BK-DEE-PALD-305
10	PAMRU	GRAM PANCHAYT	24.469189	72.402286	2022	Tube Well	ATAL JAL	BK-DEE-PAMR-306
11	PAMRU	KARSANJI DHARMAJI THAKOR	24.469389	72.402763	2022	Tube Well	ATAL JAL	BK-DEE-PAMR-307
12	PAMRU	BHURABHAI MASRUBHAI RABARI	24.468964	72.402546	2022	Tube Well	ATAL JAL	BK-DEE-PAMR-308
13	Akoli	Akoli	24.496944	72.383889	2022	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2015	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2016	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2017	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2018	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2019	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2020	Piezometer	SGWB	G_1_BK_018
	Akoli	Akoli	24.496944	72.383889	2021	Piezometer	SGWB	G_1_BK_018

S.No.	Well ID	Well Depth (mbgl)	Aquifer	PH	EC (µS/cm)	TDS (mg/L)	TH (mg/L)	Ca (mg/L)	Mg (mg/L)	Na (mg/L)	K (mg/L)	Carbonate (mg/L)	Bicarbonate (mg/L)	Sulphate (mg/L)	Chloride (mg/L)	Fluoride (mg/L)	Nitrate (mg/L)
1	BK_DAN_AAKOLI_100	121.95122	CONFINED	7.7	2760	1730.9	252	50.4	30.6	515.2	5.1	0	300	95	729.7	1.32	3.62
2	BK_DAN_AAKOLI_91	106.707317	CONFINED	7.9	2820	1764.6	166	32	20.9	568.3	5.9	0	320	13	799.7	1.3	3.5
3	BK_DAN_AAKOLI_92	91.463415	CONFINED	7.8	2950	1851.7	180	37.6	20.9	591.7	6	0	330	41	819.7	1.25	3.62
4	BK_DAN_AAKOLI_93	91.463415	CONFINED	7.8	3250	2004.7	252	52.8	29.2	627.2	6.3	0	280	74	929.6	1.39	4.25
5	BK_DAN_AAKOLI_94	182.9	CONFINED	7.7	1620	1050.1	272	52	34.5	244.4	3.9	0	310	41	359.9	1.3	3.3
6	BK_DAN_PANTHAWADA_15	228.658537	CONFINED	7.9	1000	635.7	192	38.4	23.3	140.1	1.7	0	130	88	209.9	0.99	3.22
7	BK_DAN_RAJKOT_138	91.463415	CONFINED	8.2	320	199.2	52	10.4	6.3	48.8	1.2	0	40	9	80	0.94	2.56
8	BK-DEE-PALD-304	91.463415	CONFINED	8	350	232.5	60	12.8	6.8	52	1.2	0	80	6	70	0.76	2.95
9	BK-DEE-PALD-305	83.841463	UNCONFINED	7.9	350	227.7	54	11.2	6.3	54.8	1.2	0	70	1	80	0.71	2.46
10	BK-DEE-PAMR-306	83.841463	UNCONFINED	8	400	263.3	80	14.4	10.7	54.3	1.1	0	90	9	80	0.82	3
11	BK-DEE-PAMR-307	91.463415	CONFINED	8	400	265.3	80	15.2	10.2	54.2	1.2	0	100	1	80	0.84	2.64
12	BK-DEE-PAMR-308	57.926829	UNCONFINED	8	390	251.8	100	16	14.6	42.8	1	0	90	4	80	0.8	2.69
13	G_1_BK_018	60	Unconfined_Hard Rock	8	3020	1920	346.57	25	69	535	14.45	0	427	142	720	0.92	65.4
	G_1_BK_018	60	Unconfined_Hard Rock	7.9	2460	1470	358.93	70	69	267	0	0	634	141	456	0	0
	G_1_BK_018	60	Unconfined_Hard Rock	8	2340	1480	422.26	85	51	270	5.63	0	395	120	504	2.36	23.5
	G_1_BK_018	60	Unconfined_Hard Rock	8.1	1620	980	434.36	75	60	171	5.88	0	195	71	408	1.57	7.55
	G_1_BK_018	60	Unconfined_Hard Rock	7.8	3600	2150	705.75	60	135	500	70.93	0	305	189	960	0.74	105.7
	G_1_BK_018	60	Unconfined_Hard Rock	7.9	3500	2070	582.73	80	93	532	13.6	0	183	177	1000	0.33	113
	G_1_BK_018	60	Unconfined_Hard Rock	9	2670	1650	222.9	20	42	510	6.78	24	244	130	680	1.44	1.79
	G_1_BK_018	60	Unconfined_Hard Rock	8.2	3120	1900	594.96	75	99	442	12.73	0	305	177	800	1.2	82.2

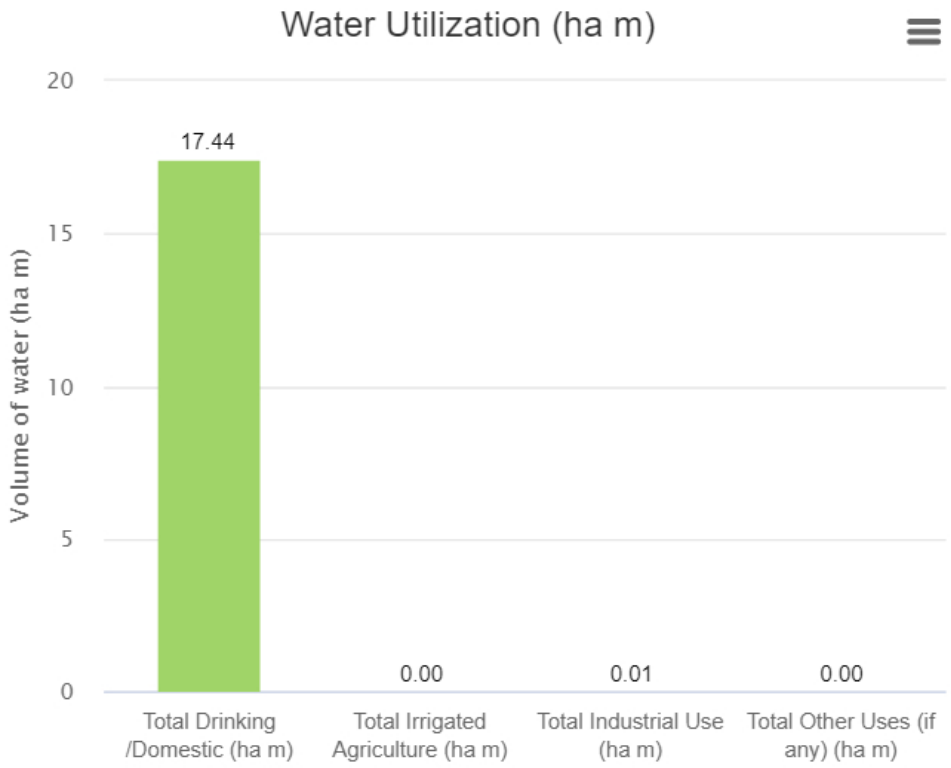
Water Availability Report

Water Availability		
1.	Total Available Water From Rainfall	
a.	Annual Rainfall (mm)	888.70
b.	Estimated Volume of Rainfall (ha m)	1681.21
c.	Evapotranspiration rate (mm)	415.01
d.	Estimated Evapotranspiration losses (ha m)	785.10
e.	Estimated Water Available From Rainfall (ha m)	896.11
f.	Weighted Average Runoff Coefficient	0.10
g.	Rainfall Runoff(ha m)	89.61
2.	Ground Water	
a.	Recharge From Rainfall (ha m)	
(i)	Monsoon	168.92
(ii)	Non-Monsoon	0.00
	Total Ground Water Recharge from rainfall (ha m)	168.92
b.	Ground Water Recharge from other sources (ha m)	
(i)	Monsoon	64.20
(ii)	Non-Monsoon	75.08
	Total Ground Water Recharge from other sources (ha m)	139.28
c.	Losses Due to Natural Discharge (ha m)	30.82
	Total Ground Water Available (ha m)	277.38
d.	Ground Water Withdrawal (ha m)	331.26
3.	Surface Water	
a.	Surface Water available in storages	
(i)	Number of Surface Water Bodies and Storage Capacity (ha m)	
Type of Structures	No. of Structures	Storage Capacity (ha m)
Ponds	2	3
	Total Storage Capacity of the surface water bodies (ha m)	3.00
(ii)	Losses if Any (%)	30.00
(iii)	Surface Water Available in Storages (ha m)	2.10
b.	Transported Water from outside GP area (ha m)	0.00
	Total Surface Water Available (ha m)	2.10
4.	TOTAL WATER AVAILABLE	
	TOTAL WATER AVAILABLE (ha m)	279.48



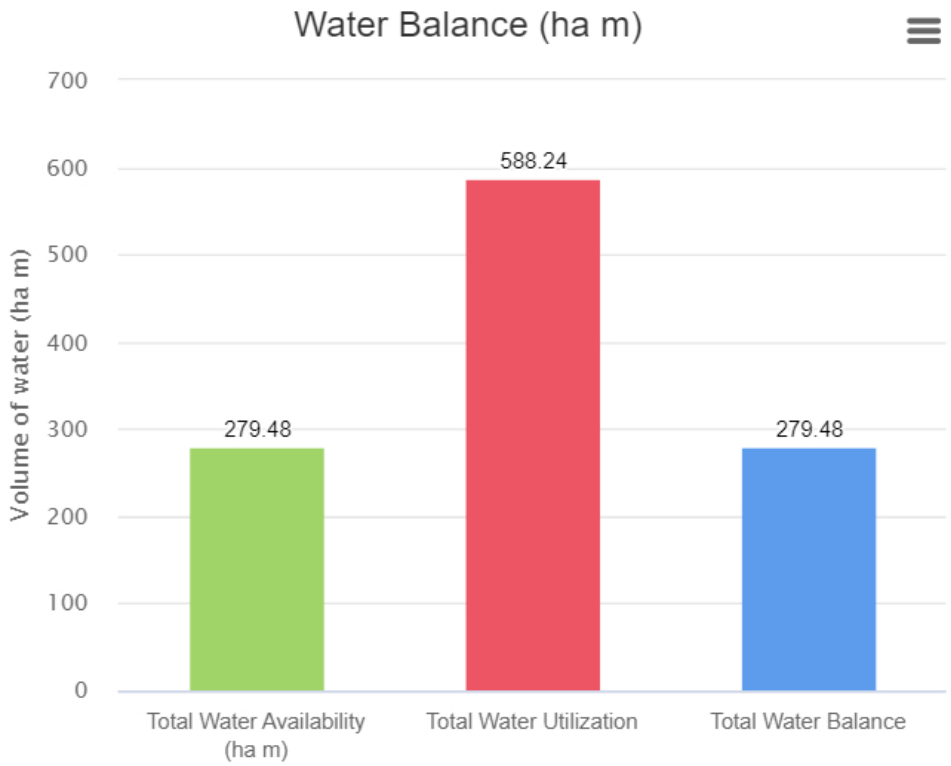
Water Utilization Report

Water Utilization		
A	Drinking /Domestic (ha m)	
i)	From Surface Water	0.79
ii)	From Ground Water	16.65
	Total Drinking /Domestic (ha m)	17.44
B	Irrigated Agriculture (ha m)	
i)	From Surface Water	0.00
ii)	From Ground Water	570.79
	Total Irrigated Agriculture (ha m)	570.79
C	Industrial Use (ha m)	
i)	From Surface Water	0.00
ii)	From Ground Water	0.01
	Total Industrial Use (ha m)	0.01
D	Other Uses (if any) (ha m)	
i)	From Surface Water	0.0000
ii)	From Ground Water	0.0000
	Total Other Uses (if any) (ha m)	0.0000



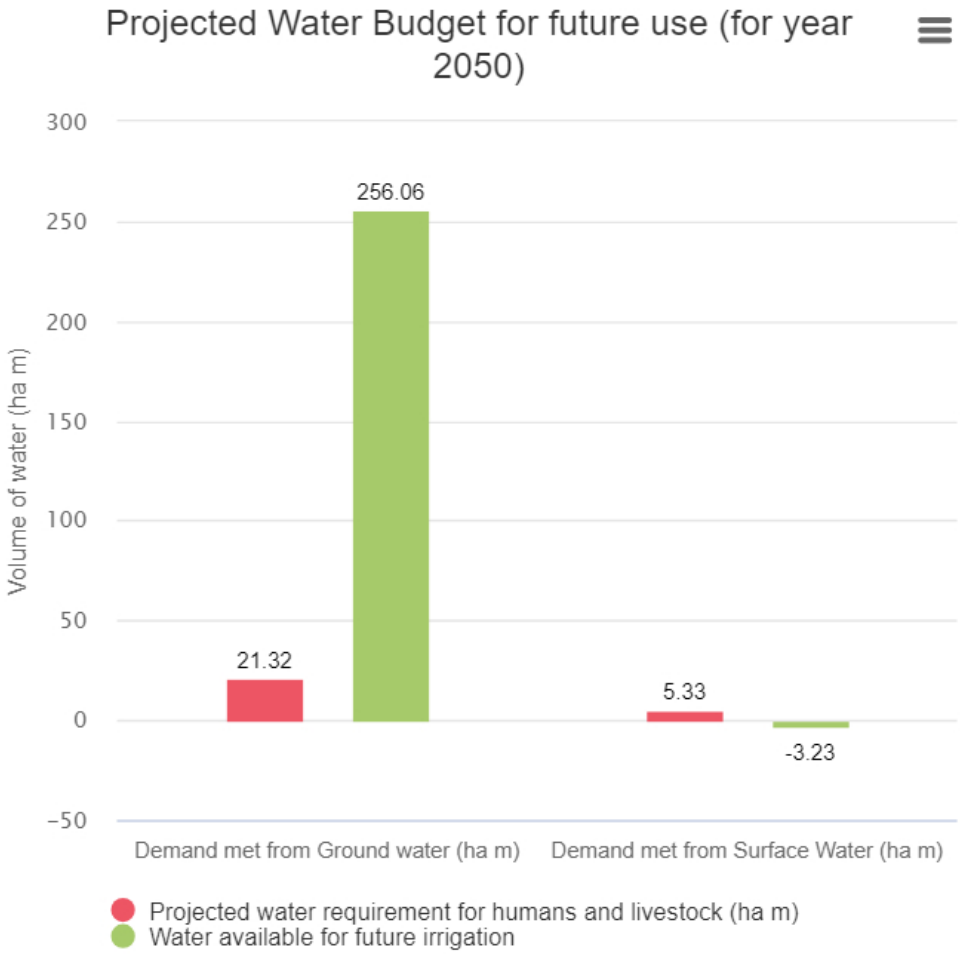
Water Balance Report

Water Balance		
A	Water Availability (ha m)	
i)	Total Ground Water Available	277.38
ii)	Total Surface Water Available	2.10
	Total Water Availability (ha m)	279.48
B	Water Utilization (ha m)	
i)	Total Surface Water Utilization	0.79
ii)	Total Ground Water Utilization	587.45
	Total Water Utilization	588.24
C	Balance [Surplus (+) / Deficit (-)]	
i)	Total Ground Water Balance	277.38
ii)	Total Surface Water Balance	2.10
iii)	Total Water Balance	279.48



Water Budget Report

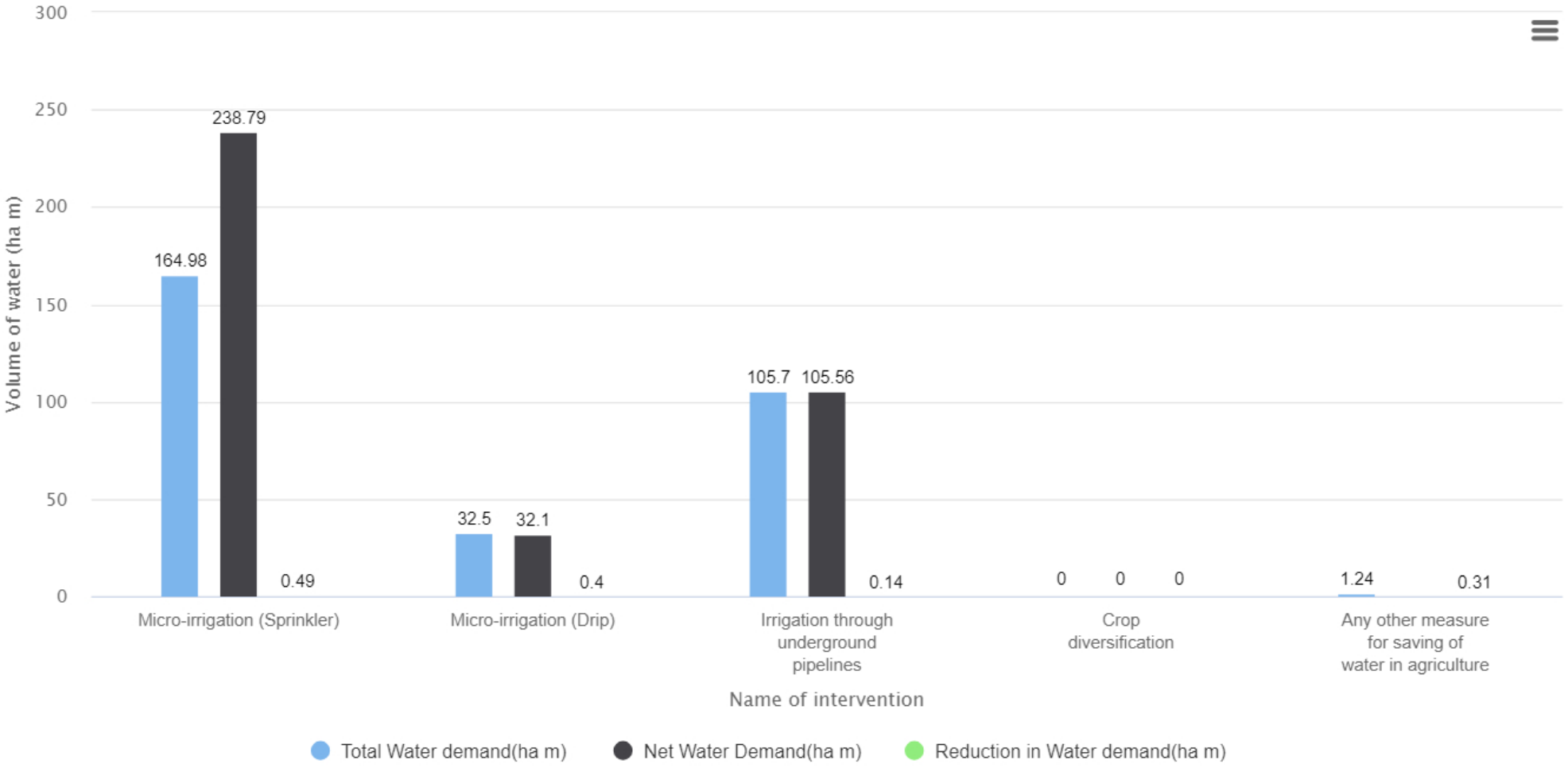
A	Allocation for Drinking & Domestic Requirements	
I	Humans	
i)	Present Human Population	3729.00
ii)	Annual population growth rate (r)	1.53
iii)	No. of years	28.00
iv)	Projected human population (as in 2050)	5705.00
v)	Per capita requirement (L)	70.00
vi)	Projected water requirement for humans (as in 2050) (ha m)	14.58
II	Livestock	
i)	Present livestock population	6930.00
ii)	Annual population growth rate (r)	1.52
iii)	No. of years	28.00
iv)	Projected livestock population (as in 2050)	10573.00
v)	Average per capita requirement (L)	31.27
vi)	Projected water requirement for livestock (as in 2050) (ha m)	12.07
B	Projected water requirement for humans and livestock (ha m) (as in 2050)	
	Total Projected water requirement for humans and livestock (as in 2050) (ha m)	26.64
i)	Proposed share to be met from Ground water (in %)	80.00
ii)	Proposed share to be met from Surface water (in %)	20.00
iii)	Demand met from Ground water (ha m)	21.32
iv)	Demand met from Surface water (ha m)	5.33
C	Water available for future irrigation	
i)	Ground Water	256.06
ii)	Surface water	-3.23
iii)	Total water available for future irrigation	252.84



Demand Side

Demand Side						
Sr No	Name of Intervention	Proposed Area(ha)	Total Water demand(ha m)	Net Water Demand(ha m)	Reduction in Water demand(ha m)	Approximate cost(in lakh)
1	Sprinkler	3.00	164.98	238.79	0.49	3.15
2	Drip	2.00	32.50	32.10	0.40	2.40
3	Pipelines	2.00	105.70	105.56	0.14	0.30
4	Crop Diversification	0	0.00	0.00	0.00	0.00
5	Other Measures	2.00	1.24	--	0.31	0.42
	Total	9.00	304.42	376.45	1.34	6.27

Funding Source						
Sr No	Name of Intervention	Area Proposed (ha)	Financial Year	Department Name	Scheme Name	Approximate Cost(in Lakh)
1	Sprinkler	3	2023-24	Gujarat Green Revolution Company Ltd. (GGRC)	Departmental Scheme	3.15
2	Drip	2	2023-24	Gujarat Green Revolution Company Ltd. (GGRC)	Departmental Scheme	2.40
3	UG Pipelines	2	2023-24	Agriculture & Farmer Welfare Department	Departmental Scheme	0.30
4	Other Measures	2	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	0.42
	Total	9				6.27

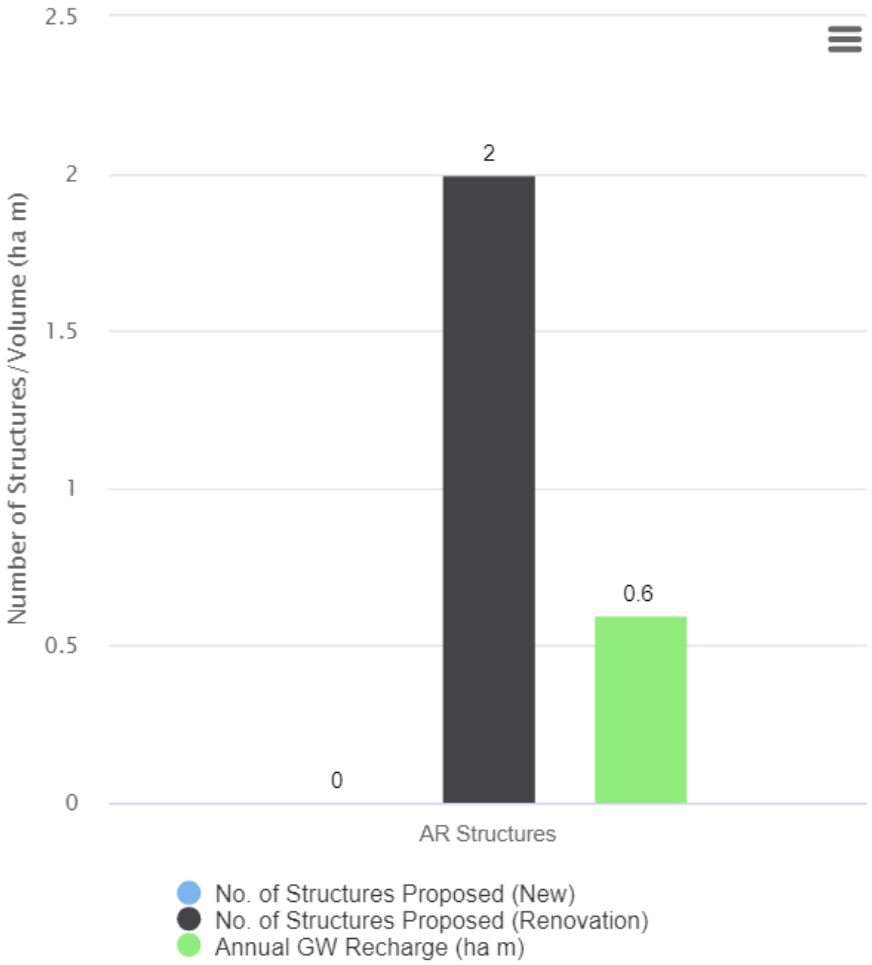

300
250
200
150
100
50
0
Volume of water (ha m)
164.98
238.79
0.49
32.5
32.1
0.4
105.7
105.56
0.14
0
0
0
1.24
0.31
Micro-irrigation (Sprinkler)
Micro-irrigation (Drip)
Irrigation through underground pipelines
Crop diversification
Any other measure for saving of water in agriculture
Name of intervention
Total Water demand(ha m)
Net Water Demand(ha m)
Reduction in Water demand(ha m)

Supply Side

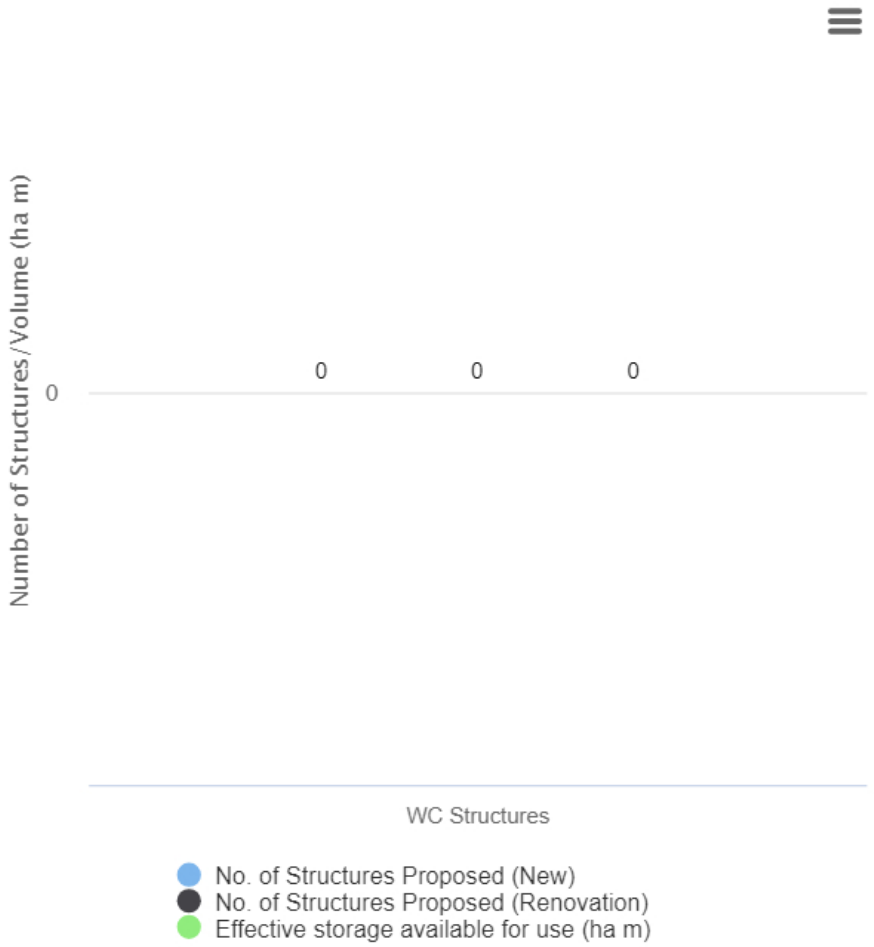
Supply Side (Artificial Recharge Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Annual GW Recharge(ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
1	DESILTING OF CHECKDAM (Renovation)	1	0.3000	Akoli	24.497605	72.38969	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	5.00
2	DESILTING OF CHECKDAM(Renovation)	1	0.3000	Akoli	24.485776	72.367663	2023-24	Gujarat Water Resources Development Corporation Ltd. (GWRDC)	Departmental Scheme	5.00
	Total	2.00	0.60							10.00

Supply Side (Water Conservation Structures)										
SR No	Name of Structure	Storage Capacity(ha m)	Effective storage available for use (ha m)	Village Name	Latitude	Longitude	Financial Year	Department Name	Scheme Name	Approximate Cost (in Lakhs)
No Data										

Supply Side (Artificial Recharge Structures)



Supply Side (Water Conservation Structures)



PGWM Committee

PGWM Committee (VWSC/WUA/CBO)				
Sr.no	Name	Designation	Gender	Mobile /Landline No.
1	KARSHANBHAI CHAMNAJE ODHAVIYA	SARPANCH	Male	9879596431
2	LERABEN POPATJEE RABARI	MEMBER	Male	8980741247
3	DABHI SARDABEN VIHABHAI	MEMBER	Male	9879465214
4	MALI ATRIEN CHHAGANBAHI	MEMBER	Female	000
5	VAGDA KASTURIBEN RUPABHAI	MEMBER	Female	7359254871
6	BHAKODRA VADLIBEN POPATBHAI	MEMBER	Female	9726701478
7	RABARI KHEMABHAI RUPAJEE	MEMBER	Male	9624801475

Details of PGWM Meeting in which resolution was passed												
Sr.no	State	District	Block	Gram Panchayat	Date	GP Name	Village Name	Meeting Location	Male	Female	Transgender	Total number of attendees
1	Gujarat_24	Banaskantha_441	Dantiwada_3735	Akoli_155943	2023-06-15	Akoli_155943	Akoli	PANCHAYT	20	1	0	21

Photograph